

A. Arduino Output - single LED

A-3. Program it to blink S.O.S. (3 short blinks, short pause, 3 long blinks, short pause, 3 short blinks, long pause, repeat).

```
/*
  Arduino SOS Blinker
  Circuit:
  - LED anode (long leg) -> digital pin 8 through a 220-330 ohm resistor
  - LED cathode (short leg) -> GND

  Morse code:
  S = ...
  O = ---
  SOS = ... --- ...
*/

const int LED_PIN = 8;

// Time values in milliseconds
const int DOT_DURATION = 250;
const int DASH_DURATION = 750;
const int SYMBOL_GAP = 250;           // Gap after each dot/dash
const int LETTER_GAP = 500;           // Additional gap between letters
const int MESSAGE_GAP = 2000;         // Long gap before repeating SOS

void setup() {
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);
}

void loop() {
  // S: three short blinks
  blinkDot();
  blinkDot();
  blinkDot();

  delay(LETTER_GAP);

  // O: three long blinks
```

```
    blinkDash();
    blinkDash();
    blinkDash();

    delay(LETTER_GAP);

    // S: three short blinks
    blinkDot();
    blinkDot();
    blinkDot();

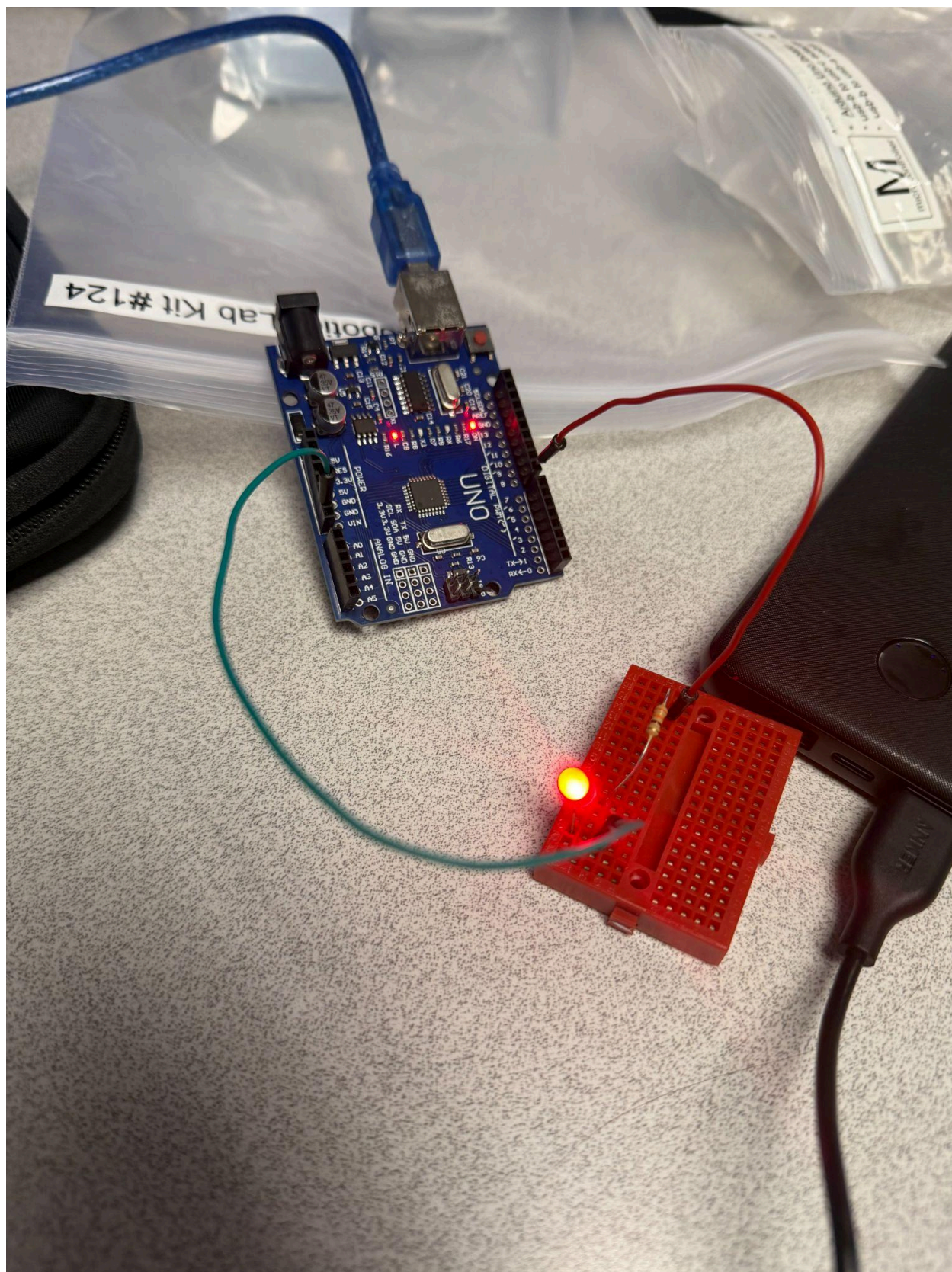
    // Long pause, then repeat the full SOS message
    delay(MESSAGE_GAP);
}

// Produces one short Morse-code signal
void blinkDot() {
    digitalWrite(LED_PIN, HIGH);
    delay(DOT_DURATION);

    digitalWrite(LED_PIN, LOW);
    delay(SYMBOL_GAP);
}

// Produces one long Morse-code signal
void blinkDash() {
    digitalWrite(LED_PIN, HIGH);
    delay(DASH_DURATION);

    digitalWrite(LED_PIN, LOW);
    delay(SYMBOL_GAP);
}
```



A-4. Write a program that will slowly bring up the brightness of the LED from completely off to fully on (0% - 100%) over the span of about 2 seconds. Have it continuously repeat the 0-100 or have it continuously cycle 0->100>0.

```
/*
  A-4: Manual PWM LED Dimmer
  -----
  Arduino Uno wiring:
    Digital pin 9 -> 220 to 330 ohm resistor -> LED anode (long leg)
    LED cathode (short leg) -> GND

  This program creates PWM manually:
    - The LED turns on for part of a short cycle.
    - The LED turns off for the rest of the cycle.
    - Increasing the ON portion increases perceived brightness.

  The program fades:
    0% -> 100% brightness in about 2 seconds
    100% -> 0% brightness in about 2 seconds
    Repeat
*/

const int ledPin = 9;

// One PWM cycle lasts 1,000 microseconds = 1 millisecond.
// That produces a PWM frequency near 1,000 Hz, fast enough
// that the LED should look steadily dim or bright.
const int pwmPeriodMicroseconds = 1000;

// Amount of time between each brightness level change.
// 8 ms x 256 brightness levels is about 2 seconds.
const int fadeStepDelay = 8;

void setup() {
  // The LED must be controlled as a digital output.
  pinMode(ledPin, OUTPUT);

  // Start with LED off.
  digitalWrite(ledPin, LOW);
}
```

```
}
```

```
void loop() {  
  // Fade from completely off (0) to fully on (255).  
  for (int brightness = 0; brightness <= 255; brightness++) {  
    runManualPWM(brightness, fadeStepDelay);  
  }  
  
  // Fade from fully on (255) to completely off (0).  
  for (int brightness = 255; brightness >= 0; brightness--) {  
    runManualPWM(brightness, fadeStepDelay);  
  }  
}
```

```
/*
```

```
  Produces manual PWM at one brightness level for a requested duration.
```

```
  brightness:
```

```
    0    = LED always off
```

```
    255 = LED always on
```

```
    1-254 = LED is on for a percentage of every PWM cycle
```

```
  durationMilliseconds:
```

```
    How long to maintain this brightness before the fade moves to  
    the next brightness value.
```

```
*/
```

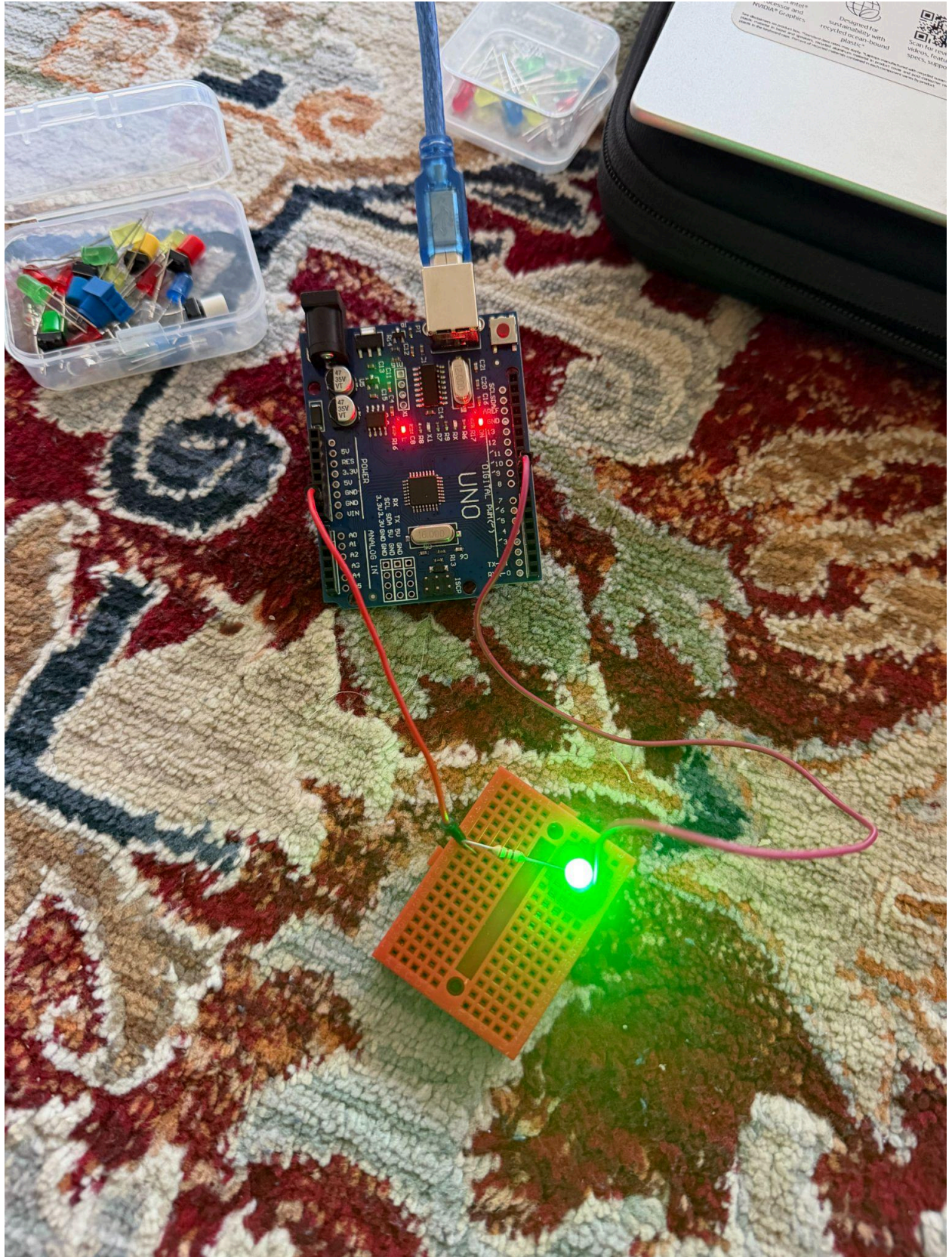
```
void runManualPWM(int brightness, int durationMilliseconds) {  
  // Convert 0-255 brightness into ON time within one PWM cycle.  
  int onTime = map(  
    brightness,  
    0,  
    255,  
    0,  
    pwmPeriodMicroseconds  
  );  
  
  // OFF time is whatever remains in the PWM cycle.  
  int offTime = pwmPeriodMicroseconds - onTime;
```

```
// Convert the requested duration into microseconds.
unsigned long totalTime =
    (unsigned long)durationMilliseconds * 1000;

// Record the time at which this brightness level begins.
unsigned long startTime = micros();

// Keep producing PWM cycles until this brightness level has lasted
// for the requested number of milliseconds.
while (micros() - startTime < totalTime) {
    // Turn LED on only if brightness is above zero.
    if (onTime > 0) {
        digitalWrite(ledPin, HIGH);
        delayMicroseconds(onTime);
    }

    // Turn LED off for the rest of the PWM cycle.
    if (offTime > 0) {
        digitalWrite(ledPin, LOW);
        delayMicroseconds(offTime);
    }
}
}
```



A-5. For this exercise, you MUST use a PWM Digital output pin and do the dim-to-bright LED with the PWM hardware control function calls.

Do a short write up on the behaviors you observed.

Here are some questions to discuss:

What is the shortest time you can pulse an LED on (with at least half a second of off time between pulses) and still be able to see it. Maybe try this in a bright room and a dark room.

Does the color of the LED make a difference on how short the pulse can be?

What if you used a smaller resistor with the LED (220 ohm) so that the LED is using more current and is brighter.

As you shorten the on and off times of your pulses the LED will blink faster. At what delay time value does the LED look like it is continuously on and not blinking?

For this exercise, I used an Arduino Uno with an external red LED connected to PWM digital pin 9. The circuit included a current-limiting resistor in series with the LED. The wiring was Arduino digital pin 9 to the resistor, then to the LED anode (long leg), with the LED cathode (short leg) connected to Arduino GND. I used `analogWrite()` to control the PWM duty cycle and gradually fade the LED from 0 to 255 brightness, then back to 0.

The LED faded smoothly from completely off to fully bright when the `analogWrite()` value increased from 0 to 255. A value near 0 produced a very dim LED because the LED was on for only a small portion of each PWM cycle. A value near 255 produced the brightest output because the LED was on for almost the entire cycle.

Shortest visible pulse:

I tested short full-brightness pulses using Arduino digital pin 9, with at least 0.5 seconds of OFF time between pulses. In a bright room, the shortest pulse I could reliably notice was approximately [enter your observed pulse length] microseconds or milliseconds. In a dark room, the shortest pulse I could reliably notice was approximately

[enter your observed pulse length] microseconds or milliseconds.

The red LED was easier to notice in the dark because there was less background light. In a bright room, the light from a short pulse had less contrast compared with the room lighting, so a longer pulse was needed before it was easy to see.

LED color:

I tested a red LED. Since I used only one LED color, I could not make a direct experimental comparison between colors. However, LED color can make a difference because LEDs can have different apparent brightness levels, and human eyes do not respond equally to every color.

Resistor comparison:

I used a 220-ohm resistor with the red LED. A 220-ohm resistor allows more current through the LED than a larger resistor, so the LED appears brighter. A brighter red LED also makes short light pulses easier to notice. The resistor is still necessary because it limits current and helps protect both the LED and the Arduino output pin.

Blinking versus continuous appearance:

I gradually shortened equal ON and OFF delays using the Arduino Uno's digital pin 9. At a delay of about [enter your observed value] ms ON and [enter your observed value] ms OFF, the LED began to appear continuously on instead of visibly blinking.

For example, if the LED looks continuous at 10 ms ON and 10 ms OFF, the full blink cycle is 20 ms. The approximate blink frequency is:

$$1000 / (10 + 10) = 50 \text{ Hz}$$

Therefore, in that example, the LED appears continuously on at about 50 Hz. My actual observed value was [enter your observed frequency] Hz.

The exact point where blinking disappears is called the flicker-fusion threshold. It can change based on LED brightness, room lighting, LED color, viewing distance, contrast, and the observer.

B-1. Write a program that will light the red, green, and yellow LED's so that they act like a stop light.

Green on for a medium time - yellow on for a short time - red on for a medium time - repeat.

**Optional*include the white LED as the walk' signal.*

```
/*
  B-1: Arduino Stoplight

  Wiring:
    D8  -> 330 ohm resistor -> red LED anode
    D9  -> 330 ohm resistor -> yellow LED anode
    D10 -> 330 ohm resistor -> green LED anode
    D12 -> 330 ohm resistor -> white LED anode (optional walk signal)

  Every LED cathode (short leg) connects to Arduino GND.

  Sequence:
    Green: 5 seconds
    Yellow: 2 seconds
    Red: 5 seconds
    Repeat
*/

// Assign each LED a descriptive pin name.
const int redLedPin = 8;
const int yellowLedPin = 9;
const int greenLedPin = 10;
const int walkLedPin = 12; // Optional white "walk" LED.

// Timing values in milliseconds.
const int mediumTime = 5000; // 5 seconds.
const int shortTime = 2000; // 2 seconds.

void setup() {
  // Configure every LED pin as an output.
  pinMode(redLedPin, OUTPUT);
  pinMode(yellowLedPin, OUTPUT);
```

```

    pinMode(greenLedPin, OUTPUT);
    pinMode(walkLedPin, OUTPUT);

    // Start with every LED off.
    allLightsOff();
}

void loop() {
    // GREEN LIGHT:
    // Cars may go. The optional walk LED is also on.
    allLightsOff();
    digitalWrite(greenLedPin, HIGH);
    digitalWrite(walkLedPin, HIGH);
    delay(mediumTime);

    // YELLOW LIGHT:
    // Warn that the signal is about to change.
    allLightsOff();
    digitalWrite(yellowLedPin, HIGH);
    delay(shortTime);

    // RED LIGHT:
    // Cars must stop. The walk LED remains off in this simple example.
    allLightsOff();
    digitalWrite(redLedPin, HIGH);
    delay(mediumTime);
}

/*
    Turns off all LEDs.

    Using this helper prevents accidental overlap, such as having
    red and green on at the same time.
*/
void allLightsOff() {
    digitalWrite(redLedPin, LOW);
    digitalWrite(yellowLedPin, LOW);
    digitalWrite(greenLedPin, LOW);
}

```

```
    digitalWrite(walkLedPin, LOW);  
}
```

B-2. Program 4 or 5 LED's so that they will continuously count in binary from 0 (0000) up to 15 (1111) and then back down to 0 again.

```
/*  
  B-2: Four-LED Binary Counter  
  
  Wiring:  
    D8  -> 330 ohm resistor -> red LED      (bit 3 / value 8)  
    D9  -> 330 ohm resistor -> yellow LED   (bit 2 / value 4)  
    D10 -> 330 ohm resistor -> green LED    (bit 1 / value 2)  
    D11 -> 330 ohm resistor -> white LED    (bit 0 / value 1)  
  
  Every LED cathode (short leg) connects to Arduino GND.  
  
  The LEDs count:  
    0 (0000) upward to 15 (1111),  
    then downward to 0 (0000),  
    and repeat.  
*/  
  
// Store the pins in order from least-significant bit to most-significant  
bit.  
// Index 0 represents bit 0 (value 1).  
// Index 3 represents bit 3 (value 8).  
const int binaryLedPins[4] = {11, 10, 9, 8};  
  
// Time to display each binary number, in milliseconds.  
const int countDelay = 750;  
  
void setup() {  
  // Configure all four LED pins as outputs.  
  for (int index = 0; index < 4; index++) {  
    pinMode(binaryLedPins[index], OUTPUT);  
  }  
}
```

```

    // Begin with 0 displayed: 0000.
    displayBinaryNumber(0);
}

void loop() {
    // Count upward from 0 (0000) to 15 (1111).
    for (int number = 0; number <= 15; number++) {
        displayBinaryNumber(number);
        delay(countDelay);
    }

    // Count back down from 14 to 0.
    // Start at 14 so that 15 is not displayed twice in a row.
    for (int number = 14; number >= 0; number--) {
        displayBinaryNumber(number);
        delay(countDelay);
    }
}

/*
Displays a number from 0 to 15 in binary using four LEDs.

The expression:
    number & (1 << bitPosition)

checks whether a particular binary bit is 1.

Examples:
    number = 5 -> binary 0101
    bit 0 = 1 -> white LED on
    bit 1 = 0 -> green LED off
    bit 2 = 1 -> yellow LED on
    bit 3 = 0 -> red LED off
*/
void displayBinaryNumber(int number) {
    // Check each of the four bit positions: 0, 1, 2, and 3.
    for (int bitPosition = 0; bitPosition < 4; bitPosition++) {
        // If the selected bit equals 1, turn its LED on.

```

```

    if (number & (1 << bitPosition)) {
        digitalWrite(binaryLedPins[bitPosition], HIGH);
    } else {
        // Otherwise, turn its LED off.
        digitalWrite(binaryLedPins[bitPosition], LOW);
    }
}
}

```

B-3. Program five LED's to act as random disco party lights

Optional Get three extra LEDs and resistors and have the LEDs spell out your name (or any string) in 7-bit ASCII.

```

/*
  B-3: Random Disco Party Lights

  Wiring:
    D8  -> 330 ohm resistor -> red LED
    D9  -> 330 ohm resistor -> yellow LED
    D10 -> 330 ohm resistor -> green LED
    D11 -> 330 ohm resistor -> blue LED
    D12 -> 330 ohm resistor -> white LED

  Every LED cathode (short leg) connects to Arduino GND.

  Important:
    Leave analog pin A0 unconnected. It is used only as a changing
    random seed when the Arduino starts.
*/

// Array containing all five LED pins.
const int discoLedPins[5] = {8, 9, 10, 11, 12};

// Number of LEDs in the array.
const int numberOfLeds = 5;

void setup() {

```

```

// Configure all LED pins as outputs.
for (int index = 0; index < numberOfLeds; index++) {
    pinMode(discoLedPins[index], OUTPUT);

    // Start with all LEDs off.
    digitalWrite(discoLedPins[index], LOW);
}

// Seed Arduino's pseudo-random generator.
// A0 should be left unconnected for this sketch.
randomSeed(analogRead(A0));
}

void loop() {
    // Choose an LED array position from 0 through 4.
    int selectedLed = random(0, numberOfLeds);

    // Randomly choose HIGH (on) or LOW (off).
    int selectedState = random(0, 2);

    // Apply the randomly selected state to the randomly selected LED.
    digitalWrite(discoLedPins[selectedLed], selectedState);

    // Random delay between 50 ms and 250 ms.
    // The upper limit is excluded by random(), so this yields 50-249 ms.
    delay(random(50, 250));
}

```

C-1. Make an Arduino sketch named serialTermDemo.ino using this code:

Lab-03 - Arduino setup, Outputs, LEDs, Serial Monitor

```

void setup() {
    // initialize both serial ports:
    Serial.begin(9600);
    Serial.print("-----\n");
    Serial.print("Hello Arduino Land");
    Serial.print("-----\n");
    pinMode(LED_BUILTIN, OUTPUT);
}

```

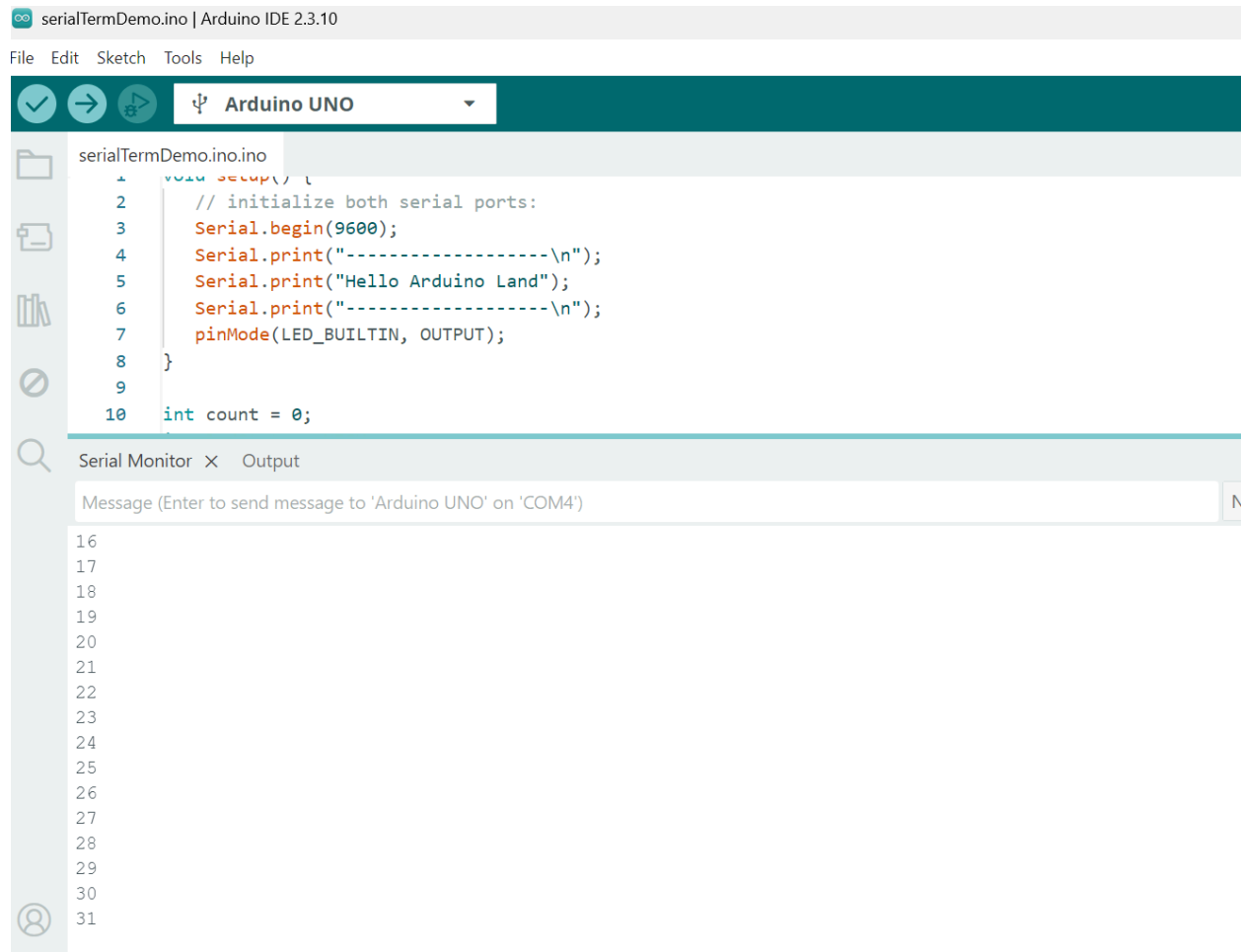
```
}

int count = 0;
int ledState = HIGH;
String countString = String(count);

void loop() {
    if(ledState == HIGH) {
        ledState = LOW;
    } else {
        ledState = HIGH;
    }

    digitalWrite(LED_BUILTIN, ledState);
    digitalWrite(LED_BUILTIN, ledState);

    Serial.println(count);
    count ++;
    delay(500);
}
```



C-3. Modify that program to make it print the count in horizontal lines of 20 numbers each with a dash between them.

/*

C-3: Serial Monitor Count in Lines of 20

This program:

- Blinks the Arduino Uno built-in LED every 500 ms.
- Prints the count to the Serial Monitor.
- Prints 20 numbers per line.
- Places a dash between every number.

Example output:

```
0 - 1 - 2 - 3 - 4 - ... - 19  
20 - 21 - 22 - 23 - ... - 39
```

```

*/

const int numbersPerLine = 20;

int count = 0;
int ledState = LOW;

void setup() {
    // Begin communication with the Serial Monitor at 9600 baud.
    Serial.begin(9600);

    // Print a title once when the Arduino starts.
    Serial.println("-----");
    Serial.println("C-3 Serial Counter");
    Serial.println("-----");

    // Configure the Uno's built-in LED as an output.
    pinMode(LED_BUILTIN, OUTPUT);
}

void loop() {
    // Toggle the built-in LED state every time loop() runs.
    if (ledState == LOW) {
        ledState = HIGH;
    } else {
        ledState = LOW;
    }

    // Apply the new LED state.
    digitalWrite(LED_BUILTIN, ledState);

    // Print the current number without moving to the next line yet.
    Serial.print(count);

    // Increase the counter after printing it.
    count++;

    /*

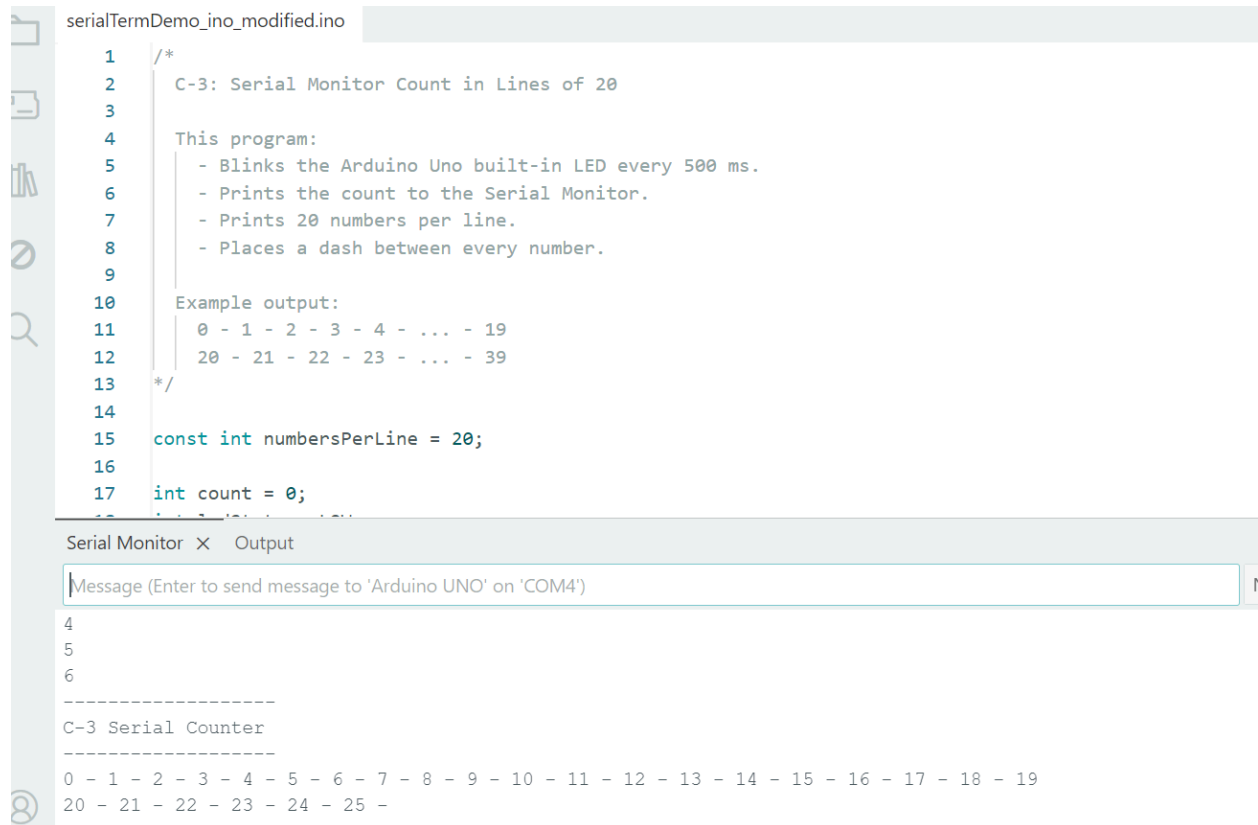
```

If 20 values have been printed, end the line.

count % 20 means "the remainder after count is divided by 20."

This condition is true after 20, 40, 60, etc. values.

```
*/  
if (count % numbersPerLine == 0) {  
    Serial.println();  
} else {  
    // Print the separator between values on the same line.  
    Serial.print(" - ");  
}  
  
// Wait half a second before printing the next number.  
delay(500);  
}
```



The screenshot shows the Arduino IDE interface. The top pane displays the code for `serialTermDemo_ino_modified.ino`. The code includes a multi-line comment describing the program's purpose (C-3: Serial Monitor Count in Lines of 20), its features (blinking the LED, printing count, printing 20 numbers per line with dashes), and example output. Below the comment, the code defines `numbersPerLine` as 20 and initializes `count` to 0. The bottom pane shows the Serial Monitor window, which has a message input field and a list of received messages. The output shows the program's execution, including the title "C-3 Serial Counter" and two lines of numbers (0-19 and 20-25) separated by dashes.

```
serialTermDemo_ino_modified.ino  
1  /*  
2    C-3: Serial Monitor Count in Lines of 20  
3  
4    This program:  
5      - Blinks the Arduino Uno built-in LED every 500 ms.  
6      - Prints the count to the Serial Monitor.  
7      - Prints 20 numbers per line.  
8      - Places a dash between every number.  
9  
10   Example output:  
11     0 - 1 - 2 - 3 - 4 - ... - 19  
12     20 - 21 - 22 - 23 - ... - 39  
13  */  
14  
15  const int numbersPerLine = 20;  
16  
17  int count = 0;  
18  
19  
20  
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```

C-4. Here is an example of an ASCII art string. Save it to a string variable and print it to the serial monitor terminal.

Find one more example and print that as well. (the tricky part is that the backslash "\" is seen as a control character. It has to be used as a pair "\\" to print out a single backslash. {<https://www.asciiart.eu/animals/cats>})

Write a program that will randomly print this example and your found (or created) ASCII art strings to the terminal, one print per second, repeat 3 times.

```
String catPic = "  ^~^  '\n ('Y\'') )\n /   \\/ \n(\\|\\|\\|/)"
```

```
/*
```

```
  C-4: Random ASCII Art to Serial Monitor
```

```
  This program:
```

- Stores two ASCII-art cat pictures in String variables.
- Randomly selects one picture.
- Prints one picture per second.
- Prints three pictures, pauses, then repeats.

```
  Important escape characters:
```

```
    \n  means print a new line.
```

```
    \\  means print one backslash character.
```

```
*/
```

```
// First ASCII cat picture from the assignment example.
```

```
// Each \\ in the code prints one \ in the Serial Monitor.
```

```
String catPicOne =
```

```
  "  ^~^  '\n"
```

```
  " ('Y') )\n"
```

```
  " /   \\/ \n"
```

```
  "(\\|\\|\\|/)"
```

```
// Second cat picture.
```

```
String catPicTwo =
```

```
  " /\\_/_/\\|\\|\\|\\n"
```

```
  "( o.o )\n"
```

```
  " > ^ <";
```

```
void setup() {
```

```
  // Start communication with the Serial Monitor.
```

```

Serial.begin(9600);

// Give the Serial Monitor a moment to open after upload.
delay(500);

// Print a heading once.
Serial.println("-----");
Serial.println("C-4 Random ASCII Cats");
Serial.println("-----");

/*
  Start the pseudo-random generator at a different location.
  A0 should be left unconnected for this program.
*/
randomSeed(analogRead(A0));
}

void loop() {
  // Print an ASCII art picture three times.
  for (int pictureCount = 1; pictureCount <= 3; pictureCount++) {
    // random(0, 2) returns either 0 or 1.
    int chosenPicture = random(0, 2);

    // Print which picture was selected.
    Serial.print("Picture ");
    Serial.print(pictureCount);
    Serial.println(":");

    // Print one of the two cats.
    if (chosenPicture == 0) {
      Serial.println(catPicOne);
    } else {
      Serial.println(catPicTwo);
    }

    // Print a blank line after each picture.
    Serial.println();

    // Wait one second before the next random picture.

```

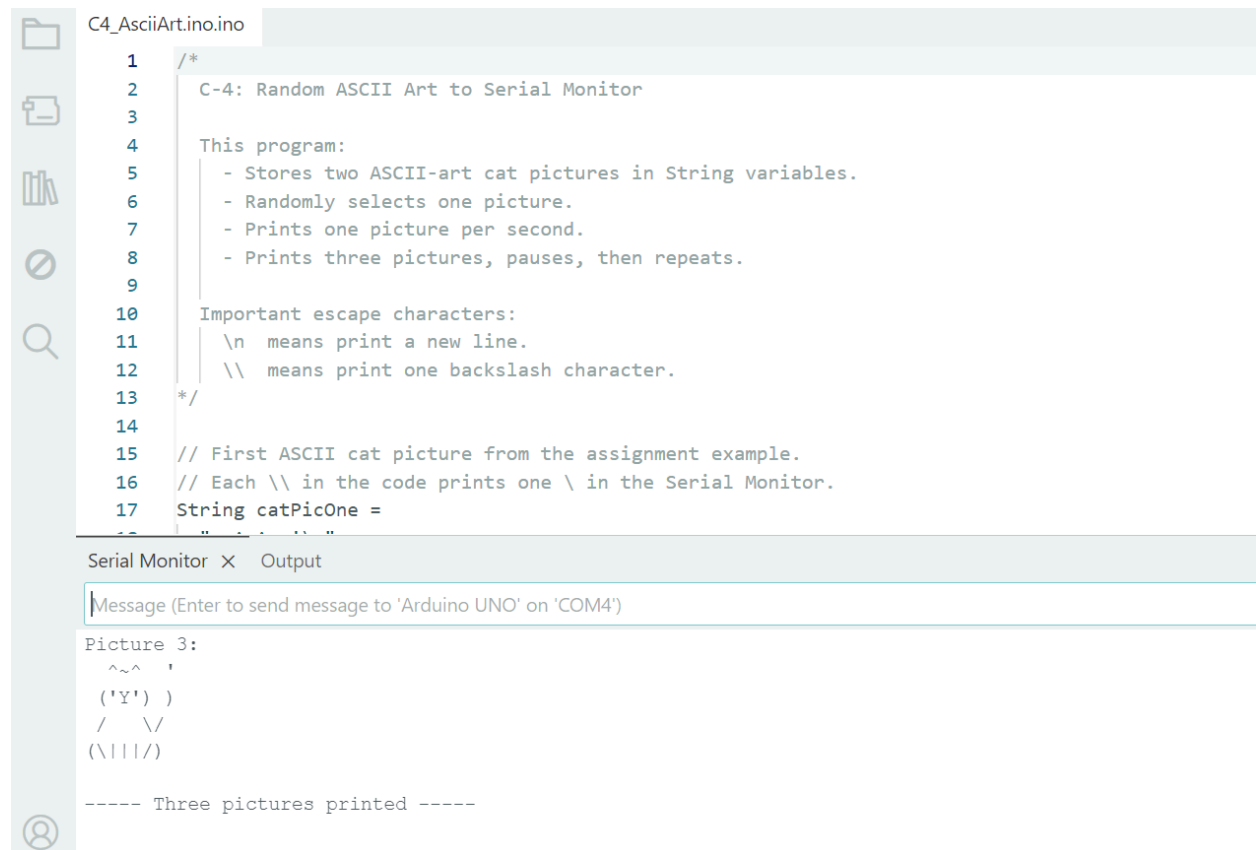
```

    delay(1000);
}

// Print an end marker after the set of three pictures.
Serial.println("----- Three pictures printed -----");
Serial.println();

// Pause before restarting the three-picture sequence.
delay(2000);
}

```



The screenshot shows the Arduino IDE interface. The top pane displays the sketch file `C4_AsciiArt.ino` with the following code:

```

1  /*
2   C-4: Random ASCII Art to Serial Monitor
3
4   This program:
5   - Stores two ASCII-art cat pictures in String variables.
6   - Randomly selects one picture.
7   - Prints one picture per second.
8   - Prints three pictures, pauses, then repeats.
9
10  Important escape characters:
11     \n means print a new line.
12     \\ means print one backslash character.
13  */
14
15  // First ASCII cat picture from the assignment example.
16  // Each \\ in the code prints one \ in the Serial Monitor.
17  String catPicOne =
18  "
19  ^~^
20  ('Y')
21  /  \
22  (\\|\\)
23  "

```

The bottom pane shows the Serial Monitor window with the following output:

```

Picture 3:
^~^
('Y')
/  \
(\\|\\)

----- Three pictures printed -----

```

C-5. Experiment with sending text from the terminal to the Arduino. Can you send it a command to turn on or off a LED?

Here is a sketch to experiment with. It will echo anything entered into the terminal monitor back to the terminal monitor. Note, the `Serial.read()` pulls in one char at a time from the terminal.

If you want a String value you will have to concatenate the chars together and `.trim()` away any stray data. (see - <https://forum.arduino.cc/t/read-line-from-serial/98251/4>)

```
// whatever you enter into the Arduino IDE serial monitor will be
echoed back to the monitor
```

```
void setup()
```

```
{
    Serial.begin(9600);
}
```

```
void loop()
```

```
{
    // Serial data is pending
    if (Serial.available())
    {
        // Echo serial data on serial device
        Serial.write( Serial.read() );
    }
}
```

```
/*
```

```
C-5: Control an Arduino LED from the Serial Monitor
```

```
This program reads text sent from the Arduino IDE Serial Monitor.
```

```
Commands to type:
```

```
ON          Turn the built-in LED on.
OFF         Turn the built-in LED off.
TOGGLE      Switch the LED between on and off.
STATUS      Print the current LED state.
HELP        Print the list of available commands.
```

```
Serial Monitor configuration:
```

```
- Baud rate: 9600
- Line ending: Newline, or Both NL & CR
```

```
*/
```

```
const int ledPin = LED_BUILTIN;
```

```
// Holds the current LED state.
```

```
bool ledIsOn = false;
```

```
// Stores characters received from the Serial Monitor until a line ends.
```

```

String incomingCommand = "";

void setup() {
    // Configure the built-in LED as an output.
    pinMode(ledPin, OUTPUT);

    // Start with the LED turned off.
    digitalWrite(ledPin, LOW);

    // Start serial communication.
    Serial.begin(9600);

    // Print instructions for the user.
    Serial.println("-----");
    Serial.println("C-5 Serial LED Controller");
    Serial.println("Type ON, OFF, TOGGLE, STATUS, or HELP.");
    Serial.println("Set Serial Monitor line ending to Newline.");
    Serial.println("-----");
}

void loop() {
    /*
        Serial.available() is greater than 0 when one or more incoming
        characters are waiting to be read.
    */
    while (Serial.available() > 0) {
        // Read one character at a time from the Serial Monitor.
        char receivedCharacter = Serial.read();

        /*
            A newline '\n' or carriage return '\r' means the user pressed
            Send, so the whole command has arrived.
        */
        if (receivedCharacter == '\n' || receivedCharacter == '\r') {
            // Remove surrounding spaces and stray line-ending characters.
            incomingCommand.trim();

            // Only process the command when something was actually typed.

```

```

        if (incomingCommand.length() > 0) {
            processCommand(incomingCommand);

            // Clear the string so it is ready for the next command.
            incomingCommand = "";
        }
    } else {
        // Add normal typed characters to the command String.
        incomingCommand += receivedCharacter;
    }
}

}

/*
    Examines a complete text command and performs the appropriate action.
*/
void processCommand(String command) {
    // Convert text to uppercase so "on", "On", and "ON" all work.
    command.toUpperCase();

    // Echo the received command back to the terminal.
    Serial.print("Received command: ");
    Serial.println(command);

    if (command == "ON") {
        // Turn LED on.
        ledIsOn = true;
        digitalWrite(ledPin, HIGH);
        Serial.println("LED is now ON.");
    }
    else if (command == "OFF") {
        // Turn LED off.
        ledIsOn = false;
        digitalWrite(ledPin, LOW);
        Serial.println("LED is now OFF.");
    }
    else if (command == "TOGGLE") {
        // Reverse the saved LED state.
        ledIsOn = !ledIsOn;
    }
}

```

```
digitalWrite(ledPin, ledIsOn ? HIGH : LOW);

if (ledIsOn) {
    Serial.println("LED toggled ON.");
} else {
    Serial.println("LED toggled OFF.");
}
}

else if (command == "STATUS") {
    // Report the current state without changing the LED.
    if (ledIsOn) {
        Serial.println("Current LED status: ON.");
    } else {
        Serial.println("Current LED status: OFF.");
    }
}

else if (command == "HELP") {
    // Show all valid commands.
    Serial.println("Available commands: ON, OFF, TOGGLE, STATUS, HELP");
}

else {
    // Handle invalid commands.
    Serial.println("Unknown command. Type HELP for valid commands.");
}

// Print a blank line to make separate commands easier to read.
Serial.println();
}
```



Arduino UNO



C5_SerialLedControl.ino.ino



```
114     if (ledison) {  
115         Serial.println("Current LED status: ON.");  
116     } else {  
117         Serial.println("Current LED status: OFF.");  
118     }  
119 }  
120 else if (command == "HELP") {  
121     // Show all valid commands.  
122     Serial.println("Available commands: ON, OFF, TOGGLE, STATUS, HELP");  
123 }  
124 else {  
125     // Handle invalid commands.  
126     Serial.println("Unknown command. Type HELP for valid commands.");  
127 }  
128  
129 // Print a blank line to make separate commands easier to read.  
130 Serial.println();  
131 }
```

Output Serial Monitor X

TOGGLE

LED is now OFF.

Received command: ON

LED is now ON.

Received command: TOGGLE

LED toggled OFF.

